

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

# Hypothesis Testing

Week 13, 14

# Hypothesis Testing

The aim is formation of a data-based decision procedure that can produce a conclusion about some scientific system. e.g.

- a) coffee drinking increases the risk of cancer in humans
- b) difference between the accuracy of two kinds of gauges
- c) person's blood type and eye color are independent variables

# Steps involved in Hypothesis Testing

- i. Define the problem
- ii. Postulate something
- iii. Make use of experimental data
- iv. Make decision based on data

A **statistical hypothesis** is an assertion or conjecture concerning one or more populations.

**Hypothesis testing** allows an experimenter to assess the plausibility or credibility of a specific statement or hypothesis.

# Steps involved in Statistical Hypothesis Testing

- i. State the Hypothesis
- ii. Test statistic
- iii. Level of Significance
- iv. Critical Region
- v. Calculations
- vi. Decision

# 1: Hypothesis

**Null hypothesis ( $H_0$ )**

**Alternative hypothesis ( $H_1$ )**

Example: Jury trial

$H_0$  : defendant is innocent,

$H_1$  : defendant is guilty

$$H_0 : \mu = \mu_0$$

$$H_A : \mu \neq \mu_0$$

# 1: Hypothesis

a. One-tailed test

$$H_0: \theta = \theta_0,$$

$$H_1: \theta > \theta_0$$

$$H_0: \theta = \theta_0,$$

$$H_1: \theta < \theta_0,$$

b. Two-tailed test

$$H_0: \theta = \theta_0$$

$$H_1: \theta \neq \theta_0$$

## 2: Test statistic

Testing of single mean (Pop variance known)

$$\begin{array}{ll} H_0: \mu = \mu_0 & \text{or} \quad H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 & H_1: \mu > \mu_0 \end{array}$$

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Test statistic does not depend on one sided or two sided hypothesis.

## 2: Test statistic

Testing of single mean (Pop variance unknown)

$$\begin{array}{ll} H_0: \mu = \mu_0 & \text{or} \quad H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 & H_1: \mu > \mu_0 \end{array}$$

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

## 2: Test statistic

Testing of two means (Pop variance known and equal)

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

## 2: Test statistic

Testing of two means (Pop variance unknown but equal)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}$$

$$s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$

## 2: Test statistic

Testing of two means (Pop variance unknown and unequal)

$$T' = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

| $H_0$                 | Value of Test Statistic                                                                                                                                                                                                              | $H_1$                                                                      | Critical Region                                                                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| $\mu = \mu_0$         | $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; \quad \sigma \text{ known}$                                                                                                                                                            | $\mu < \mu_0$<br>$\mu > \mu_0$<br>$\mu \neq \mu_0$                         | $z < -z_\alpha$<br>$z > z_\alpha$<br>$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$     |
| $\mu = \mu_0$         | $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; \quad v = n - 1,$<br>$\sigma \text{ unknown}$                                                                                                                                               | $\mu < \mu_0$<br>$\mu > \mu_0$<br>$\mu \neq \mu_0$                         | $t < -t_\alpha$<br>$t > t_\alpha$<br>$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$ | $z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}};$<br>$\sigma_1 \text{ and } \sigma_2 \text{ known}$                                                                                                | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $z < -z_\alpha$<br>$z > z_\alpha$<br>$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$ | $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}};$<br>$v = n_1 + n_2 - 2,$<br>$\sigma_1 = \sigma_2 \text{ but unknown,}$<br>$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$                       | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $t < -t_\alpha$<br>$t > t_\alpha$<br>$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$ | $t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}};$<br>$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}},$<br>$\sigma_1 \neq \sigma_2 \text{ and unknown}$ | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $t' < -t_\alpha$<br>$t' > t_\alpha$<br>$t' < -t_{\alpha/2} \text{ or } t' > t_{\alpha/2}$ |

### 3: Level of Significance

Rejection of the null hypothesis when it is true is called a **type I error**.

Non rejection of the null hypothesis when it is false is called a **type II error**.

Possible Situations for Testing a Statistical Hypothesis

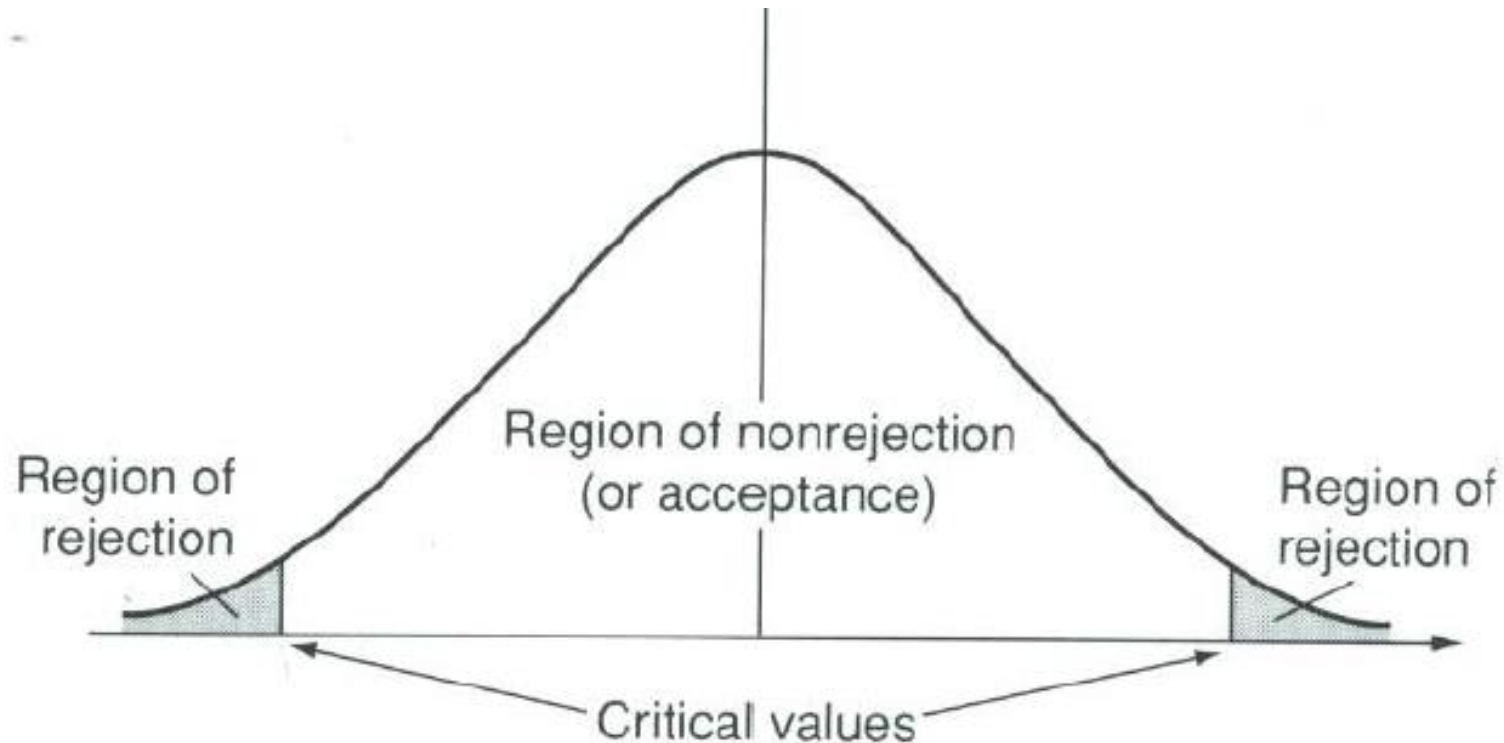
|                     | $H_0$ is true    | $H_0$ is false   |
|---------------------|------------------|------------------|
| Do not reject $H_0$ | Correct decision | Type II error    |
| Reject $H_0$        | Type I error     | Correct decision |

# 3: Level of Significance

- Probability of rejecting the null hypothesis when it is true
- $\alpha = 0.05$  means 5% risk
- Practice  $\alpha = 0.01$ ,  $\alpha = 0.05$ ,  $\alpha = 0.10$

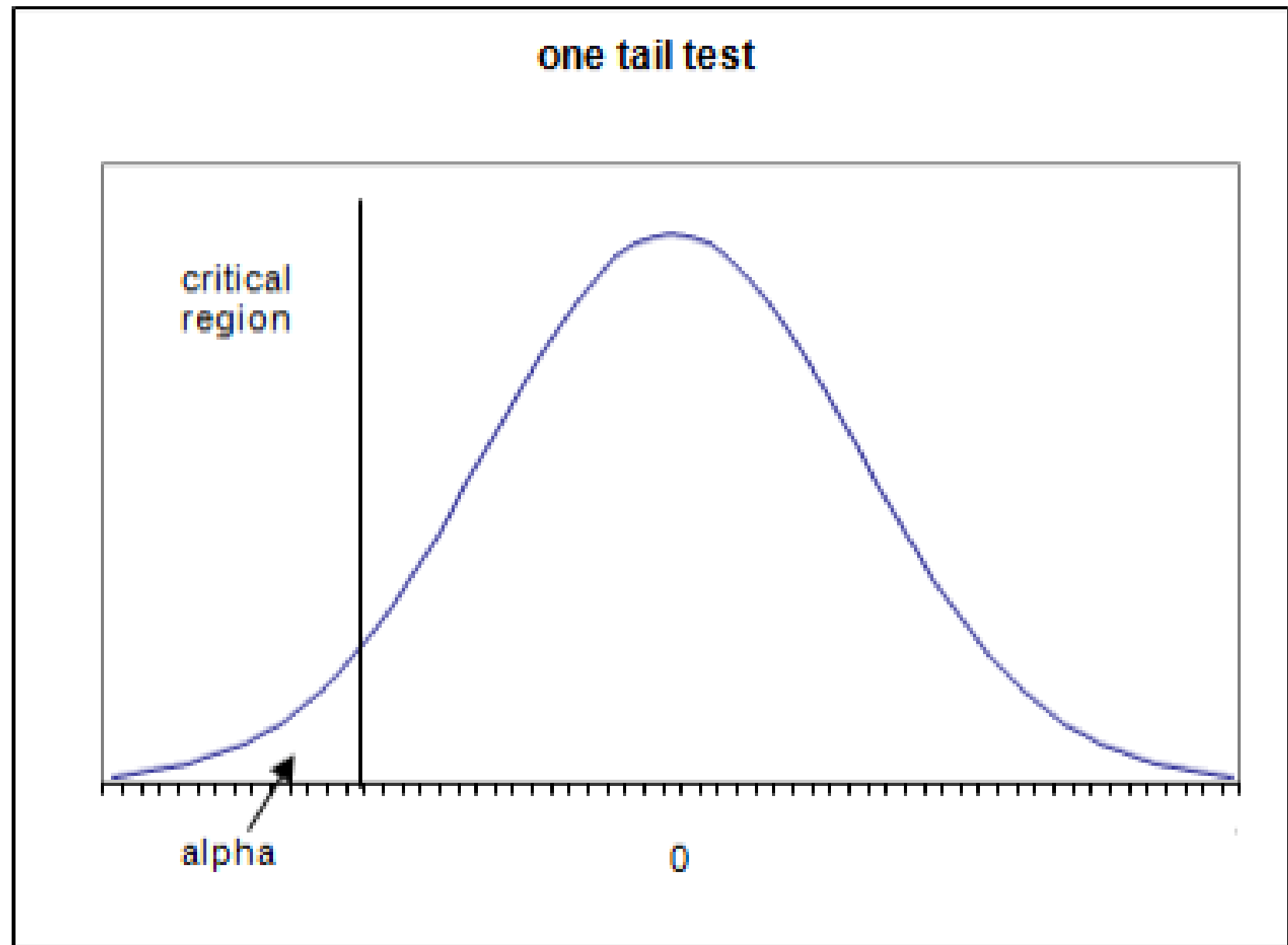
## 4: Critical Region

The critical region is the region of values that corresponds to the rejection of the null hypothesis at some chosen probability level.



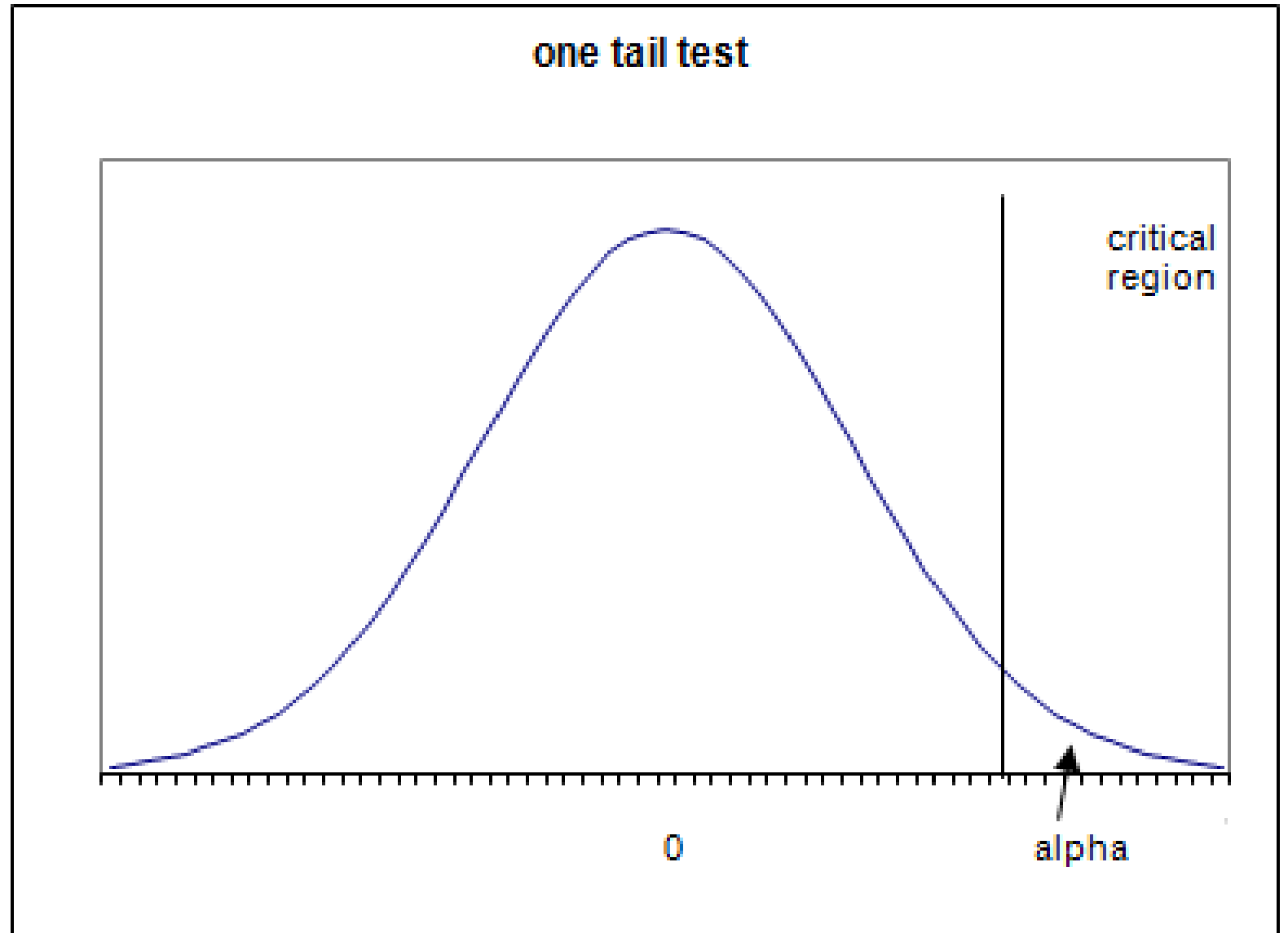
$$H_0: \theta = \theta_0,$$

$$H_1: \theta < \theta_0,$$



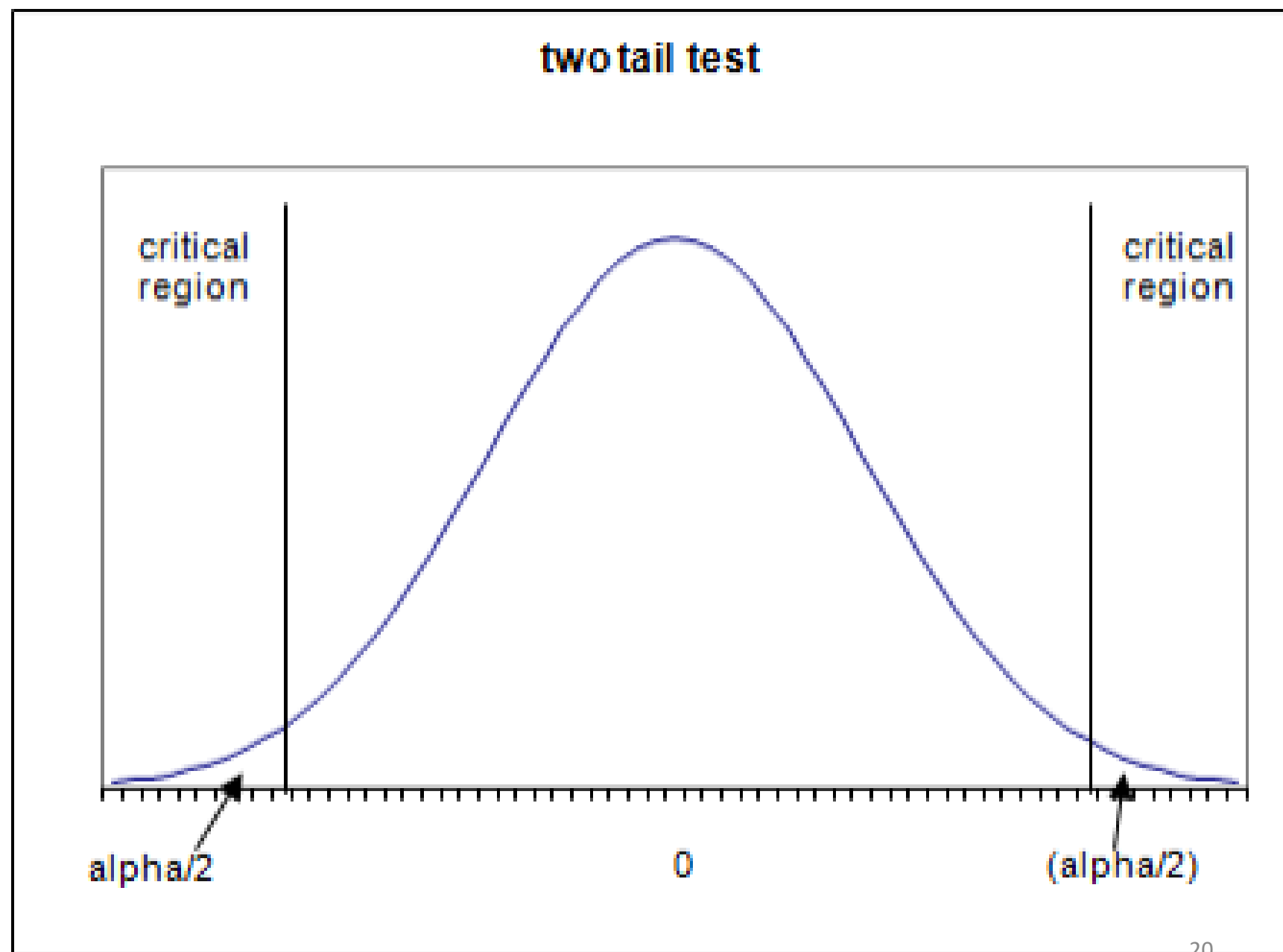
$$H_0: \theta = \theta_0,$$

$$H_1: \theta > \theta_0$$



$$H_0: \theta = \theta_0$$

$$H_1: \theta \neq \theta_0$$



## 5: Computations

## 6: Conclusion

1. Reject  $H_0$
2. Do not reject  $H_0$

Conclusions do not involve formal accept the null.

## Example: 1

A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

# Steps involved in Statistical Hypothesis Testing

- i. State the Hypothesis
- ii. Test statistic
- iii. Level of Significance
- iv. Critical Region
- v. Calculations
- vi. Decision

A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

i. Hypothesis:  $H_0: \mu = 70$  years  
 $H_1: \mu > 70$  years.

ii. Test statistic:  $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$

iii. Level of Significance:  $\alpha = 0.05$

iv. Critical Region:  $z > 1.645$

v. Calculations:



$\bar{x} = 71.8$  years,  $\sigma = 8.9$  years, and hence  $z = \frac{71.8 - 70}{8.9 / \sqrt{100}} = 2.02$

vi. Decision: Reject  $H_0$  and conclude that the mean life span today is greater than 70 years

# Task: 1

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that  $\mu = 8$  kilograms against the alternative that  $\mu \neq 8$  kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

## Example: 2

The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an **average of 46 kilowatt** hours per year. If a **random sample of 12 homes** included in a planned study indicates that vacuum cleaners use an **average of 42 kilowatt** hours per year with a **standard deviation of 11.9 kilowatt** hours, does this suggest at the **0.05 level of significance** that vacuum cleaners use, on average, **less than 46 kilowatt** hours annually? Assume the population of kilowatt hours to be normal.

i. Hypothesis:  $H_0: \mu = 46$  kilowatt hours.

$H_1: \mu < 46$  kilowatt hours.

ii. Test statistic:  $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

iii. Level of Significance:  $\alpha = 0.05$

iv. Critical Region:  $t < -1.796$

v. Calculations:  $\bar{x} = 42$  kilowatt hours,  $s = 11.9$  kilowatt hours, and  $n = 12$ .

$$t = \frac{42 - 46}{11.9/\sqrt{12}} = -1.16,$$

$$P = P(T < -1.16) \approx 0.135.$$

Decision: Do not reject  $H_0$  and conclude that the average number of kilowatt hours used annually by home vacuum cleaners is not significantly less than 46.

# P-value

A ***P*-value** is the lowest level (of significance) at which the observed value of the test statistic is significant.

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1. State null and alternative hypotheses.
  2. Choose an appropriate test statistic.
  3. Compute the *P*-value based on the computed value of the test statistic.
  4. Use judgment based on the *P*-value and knowledge of the scientific system.
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## *p*-Values

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A data set can be used to measure the plausibility of a null hypothesis  $H_0$  through the construction of a ***p*-value**. The smaller the *p*-value, the less plausible is the null hypothesis.

## Rejection of the Null Hypothesis

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A *p*-value smaller than 0.01 is generally taken to indicate that the null hypothesis  $H_0$  is not a plausible statement. The null hypothesis  $H_0$  can then be *rejected* in favor of the alternative hypothesis  $H_A$ .

## Acceptance of the Null Hypothesis

A  $p$ -value larger than 0.10 is generally taken to indicate that the null hypothesis  $H_0$  is a plausible statement. The null hypothesis  $H_0$  is therefore *accepted*. However, this does not mean that the null hypothesis  $H_0$  has been proven to be true.

## Intermediate $p$ -Values

A  $p$ -value in the range 1%–10% is generally taken to indicate that the data analysis is inconclusive. There is some evidence that the null hypothesis is not plausible, but the evidence is not overwhelming.

## Example: 3

An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume the populations to be approximately normal with equal variances.

i. Hypothesis:  $H_0: \mu_1 - \mu_2 = 2$ .

$$H_1: \mu_1 - \mu_2 > 2.$$

ii. Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}$$

iii. Level of Significance:  $\alpha = 0.05$

iv. Critical Region:  $t > 1.725$ .

$$P = P(T > 1.04) \approx 0.16.$$

v. Calculations:

$$\bar{x}_1 = 85, \quad s_1 = 4, \quad n_1 = 12.$$

$$\bar{x}_2 = 81, \quad s_2 = 5, \quad n_2 = 10.$$

$$s_p = \sqrt{\frac{(11)(16) + (9)(25)}{12 + 10 - 2}} = 4.478,$$

$$t = \frac{(85 - 81) - 2}{4.478 \sqrt{1/12 + 1/10}} = 1.04,$$

vi. Decision: Do not reject  $H_0$ . We are unable to conclude that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units.

# TASK

According to Chemical Engineering, an important property of fiber is its water absorbency. The average percent absorbency of 25 randomly selected pieces of cotton fiber was found to be 20 with a standard deviation of 1.5. A random sample of 25 pieces of acetate yielded an average percent of 12 with a standard deviation of 1.25. Is there strong evidence that the population mean percent absorbency is significantly higher for cotton fiber than for acetate? Assume that the percent absorbency is approximately normally distributed and that the population variances in percent absorbency for the two fibers are the same. Use a significance level of 0.05.

The image features a motivational quote in white text against a dark, teal-toned background. The background shows a night sky with a full moon and some silhouetted foliage at the bottom. The text is centered and reads: "Your success is measured by the strength of your desire; the size of your dream; and how you handle disappointment along the way."

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